Building AI Tutors with KIVA

MA AI Hub x SundAI — Reimagining Education with AI

Hack Day Pre-Session

MIT AI Hub

May 6, 2026

Outline

MA AI Hub x SundAI Hack Day

Mission

Explore how AI can reshape learning through hands-on prototyping

Schedule

- **Pre-session:** May 6, 6:00-7:30 PM
- **Hack Day:** May 9, 9:00 AM-5:00 PM

Who Should Join

- Educators
- Researchers
- Developers & AI builders
- Students

Focus Areas

1. **AI Tutors — Build on KIVA**
   - Evaluate tutor behavior in real scenarios
   - Prototype improvements: feedback flows, personalization, safety
   - Analyze educator interaction and intervention points
2. **Learning Path Engines**
   - Generate personalized learning pathways
   - Sequence real content (papers, courses)
   - Adapt to learner background
3. **Future of Education Systems**
   - Build educator tools
   - Create new learning experiences
   - Push education forward

KIVA: Knowledge Integration & Vocabulary Acquisition

Definition

KIVA is an **open-source AI tutoring platform** built on Riverst that enables interactive, speech-driven learning experiences with AI avatars.

Key Features

- Real-time avatar interaction
- Speech-based conversations
- Automated analysis
- Customizable flows
- Session recording

Use Cases

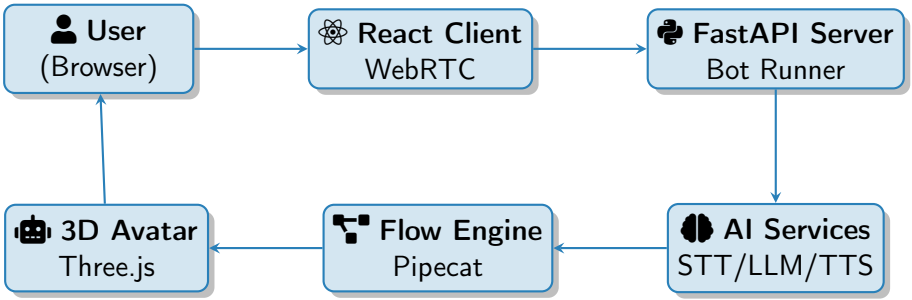
- Vocabulary tutoring
- Language learning (ESL/ISL)
- Audiobook comprehension
- Custom educational activities

KIVA Demo Videos

- **Riverst Platform Demo**
  - Overview of capabilities
  - https://drive.google.com/file/d/1r2LoBGbjBx1mdDIv-fPzGeZVEYf1kLw8/view
- **KIVA Vocabulary Activity**
  - Vocabulary teaching workflow demonstration
  - https://drive.google.com/file/d/1jTHvTXG4WWIYqgqjC4lp1LWMEnyZJQzr/view
- **Activity Summary & Configuration**
  - Configuration pages and automated analysis
  - https://drive.google.com/file/d/1AvYnSq87neOYj_YSduYatN5D4rdO-7QG/view

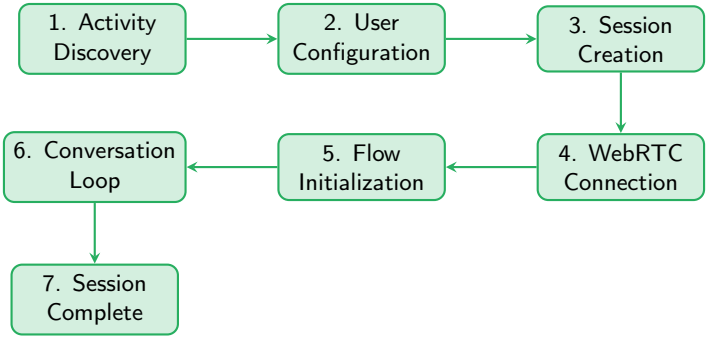
Watch these before the hack day!

How KIVA Works: High-Level Architecture



Real-time Pipeline: Speech → STT → LLM → TTS → Avatar Animation

Activity Lifecycle



- Frontend loads activities from `/api/activities`
- User configures via session schema
- Backend establishes WebRTC and initializes AI pipeline
- Flow system manages conversation state and transitions

Core Components

Frontend (React)

- `Homepage.tsx` — Activity browser
- `AvatarInteractionSettings.tsx` — Config UI
- `AvatarRenderer.tsx` — 3D avatar (Three.js)
- WebRTC connection management

Backend (FastAPI)

- `main.py` — API endpoints
- `bot_runner.py` — Orchestration
- `flow_factory.py` — Flow builder
- `handlers.py` — State management

AI Services

- **STT:** OpenAI Whisper / Local Whisper
- **LLM:** OpenAI GPT / Ollama (local)
- **TTS:** OpenAI TTS

Flow System (Pipecat)

- Stage progression tracking
- Function calling system
- Transition logic
- Context management

Analysis (Senselab)

- Speech feature extraction

System Requirements

Supported Platforms

- ✔ Apple Silicon (M-chip) macOS
- ✔ Ubuntu Linux
- ✘ Windows (not supported)

Software Requirements

- Python 3.11+
- Node.js 16+
- FFmpeg 7.*
- Modern browser (Chrome 134+)

API Keys (Optional)

- OpenAI API key
- Hugging Face token
- Google OAuth (optional)

Note

Not all API keys are required. Local models (Ollama, Whisper) can run without external APIs.

Installation: Step 1 — Clone Repository

```
# Clone the repository
git clone https://github.com/sensein/riverst.git
cd riverst
```

Repository Structure

`riverst/` contains `src/server/` (FastAPI backend), `src/client/react/` (React frontend), `KIVA/` (documentation), and `docker-compose.yaml`

Installation: Step 2 — Environment Setup

Server Configuration

- `cd src/server`
- `cp env.example .env`
- Edit .env and add your API keys

Client Configuration

- `cd src/client/react`
- `cp env.example .env`

Minimal .env for Server

OPENAI_API_KEY, HF_TOKEN, ANALYZE_AUDIO=true

Installation: Step 3 — Install Dependencies

Option A: Using Conda (Recommended)

- `conda create -n riverst python=3.11`
- `conda activate riverst`
- `conda install -c conda-forge ffmpeg`
- `pip install -r requirements.txt`

Option B: Using Docker

- `docker compose up -build`

Client setup

- `cd src/client/react && npm install`

Installation: Step 4 — Run KIVA

Start Server (Terminal 1)

- `cd src/server`
- `conda activate riverst`
- `python main.py`
- Server runs on `http://localhost:7860`

Start Client (Terminal 2)

- `cd src/client/react`
- `npm run dev`
- Client runs on `http://localhost:5173`

Important

⚠ The server **must** be running before starting the client!

Quick Start Scripts

Verification

1. Open browser to `http://localhost:5173`
2. You should see the Riverst homepage
3. Look for **"Knowledge Integration and Vocabulary Acquisition through AI (KIVA)"** section
4. Try the **"Child vocabulary training"** activity

Troubleshooting

- **Port conflicts:** Check if 7860 or 5173 are in use
- **API errors:** Verify `.env` has valid OpenAI key
- **FFmpeg missing:** Install via conda or system package manager
- **Browser issues:** Use Chrome 134+ for WebRTC support

Activity Types

Simple Activities

- Only `session_config.json`
- Open-ended conversations
- No structured flow
- Example: `basic-avatar-demo`

Advanced Flow Activities

- `session_config.json` + `flow_config.json`
- Structured conversation stages
- Dynamic content adaptation
- Example: `vocab-tutoring`

For Hack Day

You'll primarily work with **advanced flow activities** to build sophisticated tutoring behaviors.

Session Configuration

`session_config.json` defines activity parameters and UI options:

Key Configuration Fields

- AI service selection (STT/LLM/TTS)
- Avatar behavior prompts
- UI options (video, transcripts)
- Embodiment type (avatar vs waveform)

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- AI service selection (STT/LLM/TTS)
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Flow Configuration

`flow_config.json` defines conversation structure:

Flow Elements

- **Nodes:** Conversation stages (greeting, teaching, review, end)
- **Transitions:** Movement between stages based on conditions
- **Actions:** Custom Python handlers (pre/post)
- **Role messages:** System prompts for each stage

Flow Elements:

- **Nodes:** Conversation stages
- **Transitions:** Movement between stages
- **Actions:** Custom Python handlers (pre/post)

What You Can Build

Evaluation & Analysis

- Analyze tutor behavior in real scenarios
- Study conversation logs and patterns
- Identify where the AI breaks down
- Determine when human intervention is needed

Prototype Improvements

- **Feedback flows:** Better assessment and correction
- **Personalization:** Adapt to learner level and style
- **Intervention checkpoints:** When to alert educators
- **Analytics dashboards:** Track progress and engagement
- **Safety layers:** Content filtering and guardrails
- **Educator controls:** Tools for teachers to guide sessions

Example Projects

1. **Adaptive Difficulty System**
   - Monitor student performance in real-time
   - Adjust vocabulary complexity dynamically
   - Implement spaced repetition
2. **Educator Dashboard**
   - Live session monitoring
   - Intervention triggers (student struggling)
   - Post-session analytics and insights
3. **Multi-Modal Assessment**
   - Analyze speech patterns (pronunciation, fluency)
   - Track engagement via video (attention, emotion)
   - Combine metrics for holistic evaluation
4. **Safety & Content Filtering**
   - Detect inappropriate content
   - Age-appropriate language adaptation
   - Bias detection and mitigation

Technical Extension Points

Backend Extensions

- Custom handlers in `handlers.py`
- New flow configurations
- Additional AI service integrations
- Database for learner profiles
- Real-time analytics pipeline

Frontend Extensions

- New activity configuration UIs
- Live session monitoring
- Educator dashboards
- Student progress views

Flow System Extensions

- Branching logic based on performance
- Dynamic content loading
- Multi-stage assessments
- Personalization algorithms

Analysis Extensions

- Custom Senselab pipelines
- Speech quality metrics
- Engagement detection
- Learning outcome prediction

Documentation & Resources

- **GitHub Repository:** `https://github.com/sensein/riverst`
- **Activity Creation Guide:** `src/server/activities/ACTIVITY_CREATION_GUIDE.md`
- **Project Board:** `https://github.com/orgs/sensein/projects/55`
- **Pipecat Framework:** `https://github.com/pipecat-ai/pipecat`
- **Senselab:** `https://github.com/sensein/senselab`

Before Hack Day

1. Install KIVA and verify it runs
2. Watch the demo videos
3. Explore the `vocab-tutoring` activity
4. Read the Activity Creation Guide
5. Think about what you want to build!

Getting Help

During Pre-Session (May 6)

- Installation troubleshooting
- Architecture questions
- Project ideation
- Team formation

During Hack Day (May 9)

- Technical support
- Code reviews
- Integration help
- Demo preparation

Communication Channels

- Event Slack/Discord (TBD)
- GitHub Issues for bugs
- In-person mentors on hack day

Quick Reference: Key Commands

Installation

```
git clone https://github.com/sensein/riverst.git
conda create -n riverst python=3.11
conda activate riverst
pip install -r requirements.txt
```

Running

Terminal 1: `python main.py` (in src/server)
Terminal 2: `npm run dev` (in src/client/react)

Access

- Frontend: `http://localhost:5173`
- Backend API: `http://localhost:7860`

Questions?

Let's build the future of education together!

https://github.com/sensein/riverst

See you on May 9!